Privacy-Aware Decentralized Optimization

ALTERNATING DIRECTION METHOD OF MULTIPLIERS

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CS573

Outline

Beyond the Dual and Privacy Amplification

Consensus Problem:

$$\begin{aligned} \min_{\substack{x\in\mathbb{R}^{n\times p}\\ z\in\mathbb{R}^p}} \quad & \frac{1}{n}\sum_{i=1}^{n} f(x_i; d_i) + g(z) \\ \text{s.t.} \quad & x - \mathbb{I}_{n(p\times p)} z \equiv r = 0 \end{aligned} \tag{1}$$

Parameter updates in the k-th iteration:

$$z^{k+1} := \operatorname{prox}_{\gamma g} \frac{1}{n}\sum_{i=1}^{n} u_i^k \tag{2}$$

$$x_i^{k+1} := \operatorname{prox}_{\gamma f_i}(2z^{k+1} - u_i^k) \tag{3}$$

$$u_i^{k+1} := u_i^k + 2\lambda\left(r_i^{k+1} + \frac{1}{2}\eta_i^{k+1}\right) \tag{4}$$

Formulating ADMM as a Fixed-Point Iteration:

$$\text{FPI} \oplus \text{IPC} \rightarrow \text{ADMM} \;:\; u^{k+1} \equiv T_{\gamma p_1, \gamma p_2}(u^k)$$

$$\textit{Lions-Mercier Operator:} \quad T_{\gamma p_1,\gamma p_2} := \lambda(R_{\gamma p_1} \circ R_{\gamma p_2}) + (1-\lambda)I$$

$$R_{\gamma p_i} := 2\operatorname{prox}_{\gamma p_i} - I$$

Edwige Cyffers, Aurelien Bellet, and Debabrota Basu. "From Noisy Fixed-Point Iterations to Private ADMM for Centralized and Federated Learning". In: 40th International Conference on Machine Learning (Honolulu, Hawaii, USA, 2023). 2023. URL: https://proceedings.mlr.press/v202/cyffers23a/cyffers23a.pdf

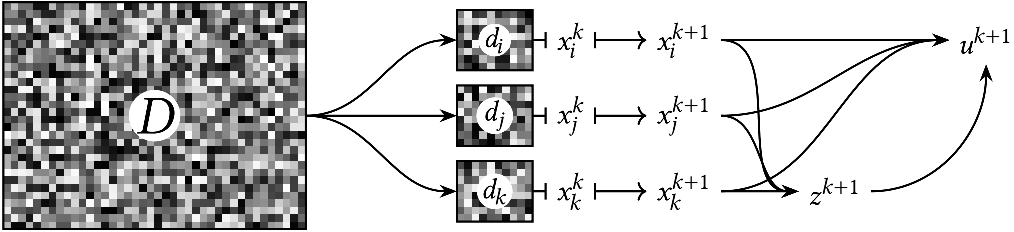
Controlled Decomposition of the Objective

Component separability and block separability

Assuming:

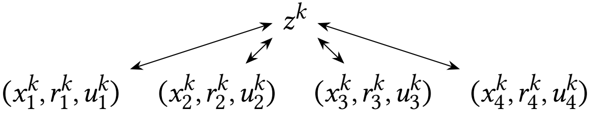
- x can be block-partitioned such that $\bigcup_i^N x_i = x$
- $f(x_0) + f(x_1) + \cdots + f(x_{N-1}) \equiv f(x)$

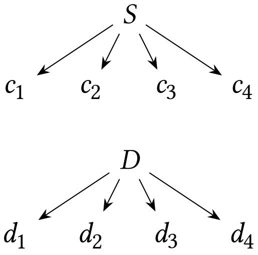
> The x-update step can be carried out *in parallel* for all $x_i \subsetneq x$.



(De)Centralization Schemes

Centralized v. Federated v. Peering





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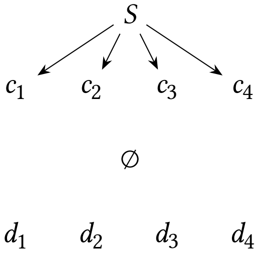
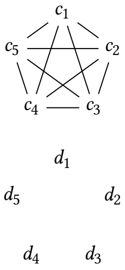


PRIVATE FEDERATED



PEERING

An Acceptable Federated Implementation

Objective: Empirical Risk Minimization with Federated RDP-FPI-ADMM

$$\min_{x\in\mathbb{R}^d} \mathbb{E}_{d_i\sim D}\left[f_i(x,d_i)\right] + g(x) \qquad \min_{\substack{x\in\mathbb{R}^{n\times p}\\ z\in\mathbb{R}^p}} \{\mathbb{E}_{\sim(x,D)} f(x_i;d_i) + g(z) \mid x = \mathbb{I}_{n(p\times p)} z\}$$

1. Construct the federated implementation of ADMM (e.g. OpenMPI or simple sockets).
2. Establish privacy guarantees (robustness to reconstruction attacks) by leveraging the Rényi Differential Privacy (RDP) (ε,δ)-formulation.
3. Stabilize optimization by mitigating client drift in heterogeneous (non i.i.d) regimes. Benchmark with various degrees of heterogeneity.
4. Implement resource-aware aggregation as opposed to a simple averaging.

Outline

Simulating a Diverse Set of Clients

In federated learning, client datasets may differ in both *mode* and *size*.

- Finding such datasets would be highly non-trivial.
- ***Instead, can we effectively simulate this?***

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Splitting a Dataset in a Biased Way

1. Split a dataset into a number of *modes*, one for each client.
2. Sample with *high probability* (many points) from the *modal range*.
3. Sample with *low probability* (few points) from the *residual range*, i.e. outside the modal range.
4. Randomize the points sampled to make the datasets differ in size.

Non-IID Data Splitting for Regression Datasets

Function RandPop(Array, End, NPoints)

Input: An array Array with dim Array $\equiv n \times m$, a right range limit End and the number of points to sample NPoints.

Output: A *quasi*-random subsample Sample of Array and Array with the elements of Sample removed.

```
assert {End, NPoints} ∈ ℕ
assert 0 ≤ NPoints ≤ End < n
Sample ← uniformly sample NPoints from the range Array[0 : End]
Array ⟵ Array \ Sample
return Sample, Array
```

Non-IID Data Splitting for Regression Datasets

Input: A Dataset Data, the column-index of the target TgtIdx, the number of splits to produce Splits, a "bias factor" Bias and a size-randomization toggle RandSizes.

Output: A Splits-long hashmap Hmap of biased sub-datasets.

Data ← `SortAscBy(`Data; TgtIdx`)`   `/* sort data by target values */`

for $0 \leq i <$ Splits **do**

    if RandSizes *is* `True` **then**   `/*` rand $\in [0.5, 1] \vee 1$ `*/`

        rand $\leftarrow 1 + \mathcal{N}(0,1)/2$

    else rand $\leftarrow 1$

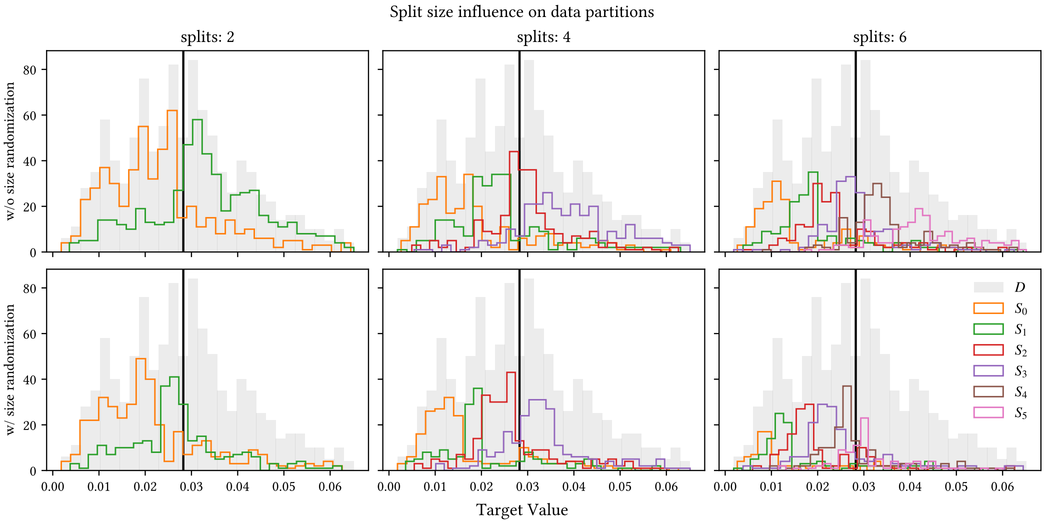
    nsplit $\longleftarrow \lfloor$rand $\cdot$ `Len(`Data`)`/Splits$\rfloor$

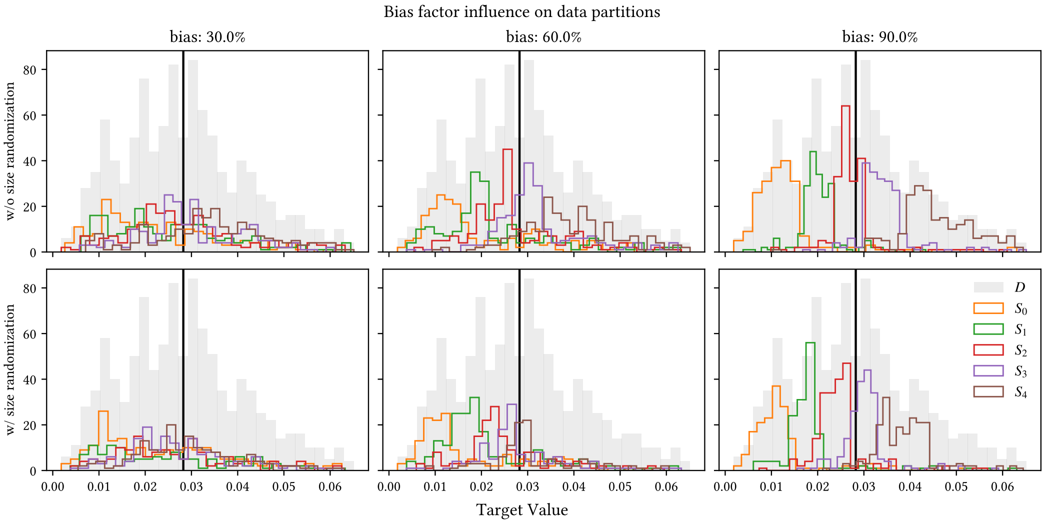
    nmode $\longleftarrow \lfloor$Bias $\cdot$ nsplit$\rfloor$

    nrsd $\longleftarrow$ nsplit $-$ nmode

    Hmap[i] $\leftarrow$ `RandPop(`Data, `Len(`Data`)`, nrsd`)` :: `RandPop(`Data, nsplit, nmode`)`

return Hmap


Split size influence on data partitions
splits: 2
splits: 4
splits: 6
w/o size randomization
w/ size randomization
0
20
40
60
80
0.00
0.01
0.02
0.03
0.04
0.05
0.06
Target Value
D
S_0
S_1
S_2
S_3
S_4
S_5


Bias factor influence on data partitions
bias: 30.0%
bias: 60.0%
bias: 90.0%
w/o size randomization
w/ size randomization
0
20
40
60
80
0.00
0.01
0.02
0.03
0.04
0.05
0.06
Target Value
D
S0
S1
S2
S3
S4

Outline

Offering Privacy Guarantees

> Client data may be *sensitive* or *confidential*. ***How can we effectively inform them of their potential exposure?***

Cynthia Dwork and Aaron Roth. "The Algorithmic Foundations of Differential Privacy". In: vol. 9. Foundations and Trends in Theoretical Computer Science 3 - 4. 2014. URL: https://www.cis.upenn.edu/~aaroth/Papers/privacybook.pdf.

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Hospitals want to contribute to clinical trials *without publishing patient records*.

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Reconstruction Attacks

For a database D with n records, query Q and a response R let $\mathscr{M}_D : Q \mapsto R$ be a mechanism.

$$\text{[Dwork 2014]}\ \frac{\exists A : \langle \mathscr{M}(Q_1), \mathscr{M}(Q_2), \ldots, \mathscr{M}(Q_m) \rangle \mapsto D' \ni \|D' - D\|_1 \in o(n)}{\mathscr{M} \text{ is } \textit{blatantly non-private}}$$

Cynthia Dwork and Aaron Roth. "The Algorithmic Foundations of Differential Privacy". In: vol. 9. Foundations and Trends in Theoretical Computer Science 3 - 4. 2014. URL: https://www.cis.upenn.edu/~aaroth/Papers/privacybook.pdf.

Pure Differential Privacy

Definition (Differential Privacy)

A randomized mechanism $\mathscr{M} \in \mathbb{N}^{|\mathcal{X}|}$ is (ϵ, δ)-differentially private iff:

$$\|D - D'\|_1 \leq 1, \mathcal{S} \subseteq \operatorname{ran} \mathscr{M} \vdash \Pr[\mathscr{M}_D \in \mathcal{S}] \leq e^{\epsilon} \Pr[\mathscr{M}_{D'} \in \mathcal{S}] + \delta$$

$$\text{Abs. Privacy Loss:} \qquad \left| \ln \frac{\Pr[\mathscr{M}_D = \xi]}{\Pr[\mathscr{M}_{D'} = \xi]} \right| \leq \epsilon \quad \textit{with probability } 1 - \delta$$

δ characterizes *catastrophic failure* and must ideally be ***negligible***.

Cynthia Dwork and Aaron Roth. "The Algorithmic Foundations of Differential Privacy". In: vol. 9. Foundations and Trends in Theoretical Computer Science 3 - 4. 2014. URL: `https://www.cis.upenn.edu/~aaroth/Papers/privacybook.pdf`.

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Max Divergence: $$D_{\infty}^{\delta}(\mathscr{M}_D \| \mathscr{M}_{D'}) := \max_{\mathcal{S} : \Pr[D' \in \mathcal{S}] \geq \delta} \ln \frac{\Pr[\mathscr{M}_D \in S] - \delta}{\Pr[\mathscr{M}_{D'} \in S]} \leq \epsilon$$

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Necessary Properties of DP Mechanisms

- Protection against arbitrary risks.
- Neutralization of linkage attacks.
- Quantification of privacy loss.
- Composition.
- Closure under post-processing.

Cynthia Dwork and Aaron Roth. "The Algorithmic Foundations of Differential Privacy". In: vol. 9. Foundations and Trends in Theoretical Computer Science 3 - 4. 2014. URL: https://www.cis.upenn.edu/~aaroth/Papers/privacybook.pdf.

Adding Noise to the Responses

> For some function f define a randomized mechanism $\mathscr{M} \ : \ f(x) + \textit{some noise}.$

Laplace Mechanism $\mathscr{M}_{\mathrm{L}}[x, f; \lambda] \coloneqq f(x) + \Lambda(0, \lambda),\ \lambda \propto \Delta^p f / \epsilon.$

Gaussian Mechanism $\mathscr{M}_{\mathrm{G}}[x, f; \sigma^2] \coloneqq f(x) + \mathcal{N}(0, \sigma^2),\ \sigma^2 \propto \Delta^p f / \epsilon.$

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> Under the Gaussian Mechanism
>
> - The noise of the dataset is also normally distributed.
> - The tails of the Gaussian distribution decay much faster than the tails of the Laplace one.
> - Cannot meet ϵ-DP for all ϵ.

Reformulation based on Rényi Divergence

Definition (Rényi Divergence)

For two probability distributions P and Q, the Rényi divergence of order $\alpha > 1$ is:

$$D_\alpha(P\|Q) := \frac{1}{\alpha - 1} \ln \mathbb{E}_{x\sim Q}\left[\frac{P(x)}{Q(x)}\right]^\alpha$$

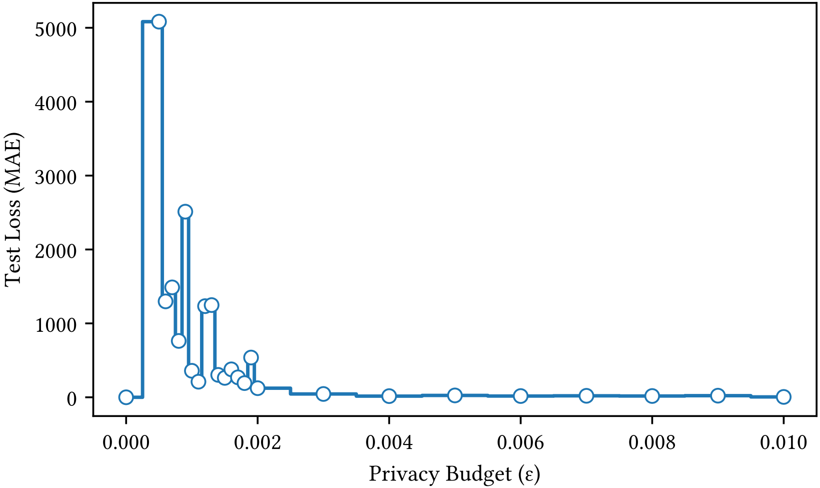
(α, ε)-Rényi Differential Privacy

$$D_\alpha(\mathcal{M}_D\|\mathcal{M}_{D'}) \leq \varepsilon$$

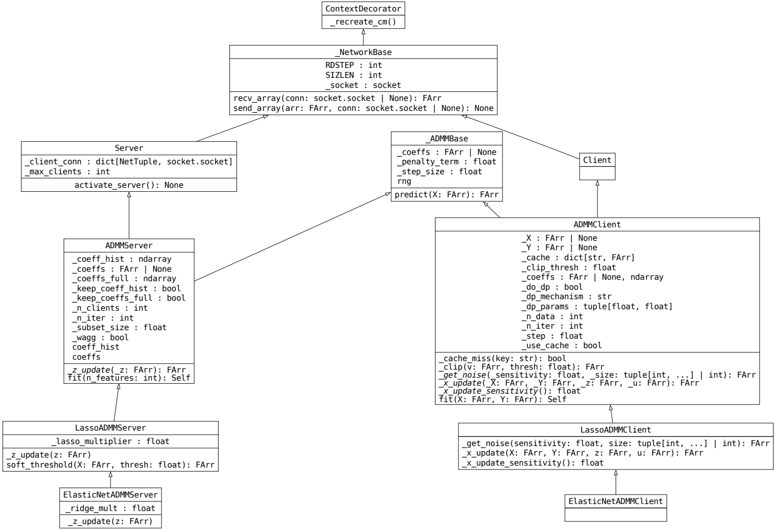
$D_\alpha(P\|Q) \leq D_\infty(P\|Q) \vDash$ *An (α, ε)-RDP mechanism also satisfies $\left(\varepsilon + \frac{\ln 1/\delta}{\alpha-1}, \delta\right)$-DP.*

Ilya Mironov. "Rényi Differential Privacy". In: IEEE 30th Computer Security Foundations Symposium (2017). 2017. DOI: 10.1109/CSF.2017.11.

Performance Impact of Privacy Budget



Outline


ContextDecorator
_recreate_cm()
_NetworkBase
RDSTEP : int
SIZLEN : int
_socket : socket
recv_array(conn: socket.socket | None): FArr
send_array(arr: FArr, conn: socket.socket | None): None
Server
_client_conn : dict[NetTuple, socket.socket]
_max_clients : int
activate_server(): None
_ADMMBase
_coeffs : FArr | None
_penalty_term : float
_step_size : float
rng
predict(X: FArr): FArr
Client
ADMMServer
_coeff_hist : ndarray
_coeffs : FArr | None
_coeffs_full : ndarray
_keep_coeff_hist : bool
_keep_coeffs_full : bool
_n_clients : int
_n_iter : int
_subset_size : float
_wagg : bool
coeff_hist
coeffs
_z_update(_z: FArr): FArr
fit(n_features: int): Self
ADMMClient
_X : FArr | None
_Y : FArr | None
_cache : dict[str, FArr]
_clip_thresh : float
_coeffs : FArr | None, ndarray
_do_dp : bool
_dp_mechanism : str
_dp_params : tuple[float, float]
_n_data : int
_n_iter : int
_step : float
_use_cache : bool
_cache_miss(key: str): bool
_clip(v: FArr, thresh: float): FArr
_get_noise(_sensitivity: float, _size: tuple[int, ...] | int): FArr
_x_update(_X: FArr, _Y: FArr, _z: FArr, _u: FArr): FArr
_x_update_sensitivity(): float
fit(X: FArr, Y: FArr): Self
LassoADMMServer
_lasso_multiplier : float
_z_update(z: FArr)
soft_threshold(X: FArr, thresh: float): FArr
LassoADMMClient
_get_noise(sensitivity: float, size: tuple[int, ...] | int): FArr
_x_update(X: FArr, Y: FArr, z: FArr, u: FArr): FArr
_x_update_sensitivity(): float
ElasticNetADMMServer
_ridge_mult : float
_z_update(z: FArr)
ElasticNetADMMClient

Server Loop

```
Class Server(ElasticNet, ADMM, Net)
    Function fit(NFeat)
        foreach k in clients do
            generate weight w_k
        for 0 ≤ i < NIter do
            subset ← sample from clients
            foreach conn in subset do
                Self.send(conn, z)
            while Len(subset) > 0 do
                u_update(subset, du)
            du ← du/Len(clients)
            z ← Self.z_update(du)
        return Self
```

```
Class Server(ElasticNet, ADMM, Net)
    Function Server(...)
        return Self w/ parameters initialized

    Function u_update(subset, du)
        foreach conn in select(subset) do
            u ←—— Self.recv(conn)
            du ←—— du + w_k u
            subset ← subset \ conn

    Function z_update(z)
        return 1/(1+2ρλ_2) S_{λ_1/ρK}(z)
```

$$\frac{1}{1+2\rho\lambda_2} S_{\lambda_1/\rho K}(z)$$

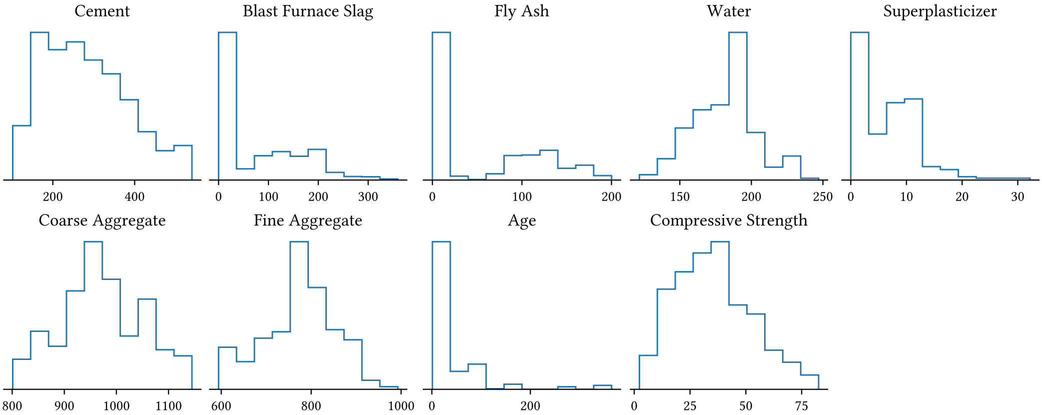
Client Loop

Class `Client(ElasticNet, ADMM, Net)`
- **Function** `fit(NFeat)`
  - *send server DP-weight data*
  - **for** $0 \leq i <$ NIter **do**
    - $z \leftarrow$ Self.recv()
    - $x \leftarrow$ Self.x_update(X, Y, z, u)
    - $r \leftarrow x - z + \mathcal{N}(\Delta^2 f/(2\varepsilon)^2)$
    - $\mathrm{d}u \leftarrow 2\lambda r$
    - Self.send($\mathrm{d}u$)
    - $u \leftarrow u + \mathrm{d}u$
  - **return** Self

Class `Client(ElasticNet, ADMM, Net)`
- **Function** `Client(...)`
  - *connect to server*
  - *initialize cache*
  - **return** Self *w/ parameters initialized*
- **Function** `x_update(`X, Y, z, u`)`
  - $l \leftarrow \rho/2$
  - lhs $\leftarrow$ Self.cache($X^\top X + l\mathbb{I}$)
  - xty $\leftarrow$ Self.cache($X^\top Y$)
  - **return** lhs $\cdot$ (xty $+ l(z - u)$)

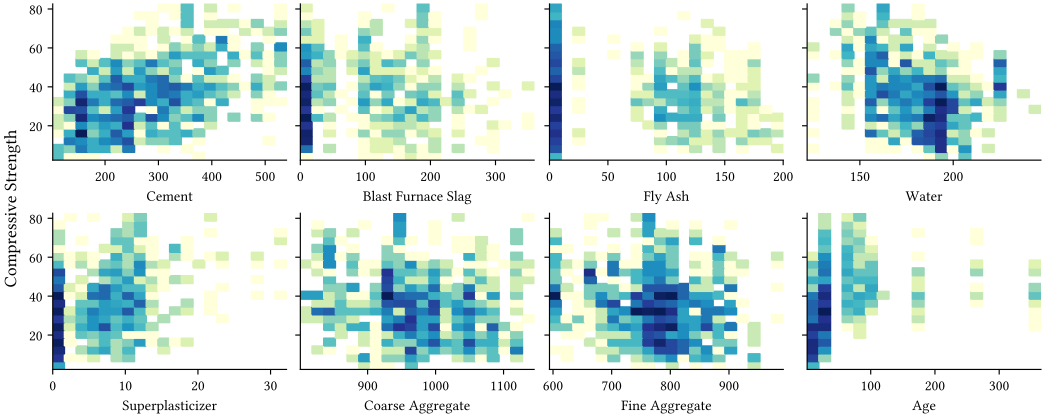
The Concrete Dataset

Feature Distributions



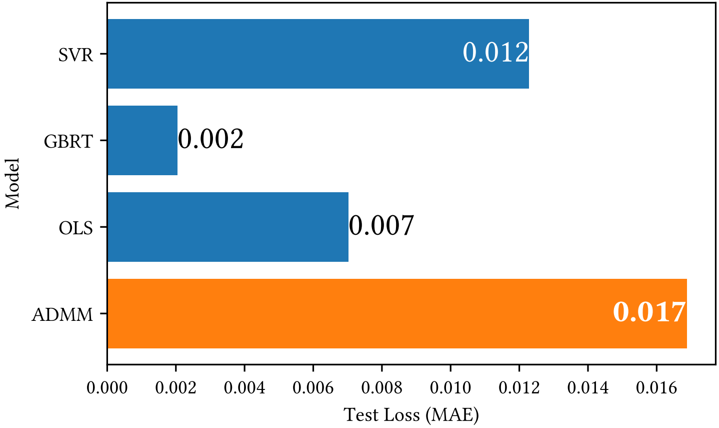
The Concrete Dataset

Feature–Target Correlation



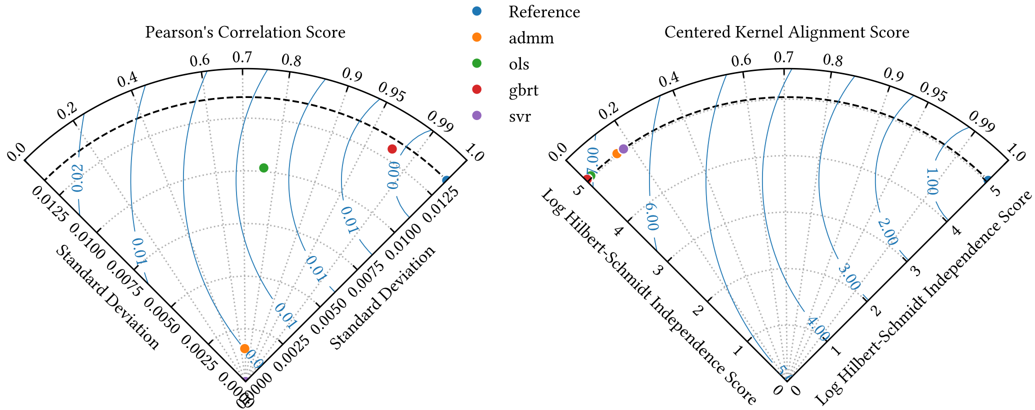
Performance

MAE Scores



Performance

Taylor Diagrams



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- Eric Chu Borja Peleato Stephen Boyd Neal Parikh and Jonathan Eckstein. "Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers". In: vol. 3. Foundations and Trends in Machine Learning 1. 2011, pp. 1–122. URL: https://web.stanford.edu/~boyd/papers/pdf/admm_distr_stats.pdf.